

Phase 3: Project Design

AI Comic Story Creator — SmartBridge Project Submission

1. High-Level Architecture

The system is a decoupled, multi-agent pipeline. User Prompt → Gemini 1.5 Pro Script Master (narrative, beats, characters) → Character Consistency Engine (visual seeds & style tokens) → Gemini Layout & Panel Agent (composition & bubble anchor) → Imagen 3 Image Generation (diffusion panel synthesis) → Typesetting & Compositor (PDF/booklet export engine).

2. Agent Specifications

Agent	Core Model	Function
1. Story Architect	Gemini 1.5 Pro	Generates a 7-8 page script with narrative arcs, page breaks, panel beats
2. Character Manager	Gemini 1.5 Flash	Generates static visual anchor descriptors enforced across panels
3. Panel Director	Gemini 1.5 Flash	Produces image prompts: camera angle, lighting, background, bubble anchors
4. Image Renderer	Imagen 3 / Ultra	Synthesizes artwork in the specified style (Manga, Silver Age, Noir, etc.)
5. Typesetter/Publisher	Canvas API / Pillow	Draws bubbles & captions, compiles final PDF/CBZ pages

3. Data Schema Design

All inter-agent communication uses strict JSON schemas (enforced via Gemini's response_schema parameter) so downstream agents can rely on predictable structure. The script object includes: project_title, total_pages, art_style, a characters array (each with an id, name, and visual_anchor string), and a pages array (each with panel_count and a panels array of camera_angle, environment, action_description, and a dialogue array of speaker/type/text/position).

4. Visual Consistency Design

Since no native model fine-tuning is used, consistency is enforced structurally: every image prompt sent to Imagen 3 is built by concatenating [ART STYLE PREFIX] + [CHARACTER VISUAL ANCHOR] + [PANEL ACTION] + [ENVIRONMENT & LIGHTING]. The same character's visual anchor string is reused verbatim on every panel in which that character appears.

5. Page Layout Design

A layout engine maps panel counts to grid templates (1-panel splash, 4-panel standard, 3-panel dynamic, and additional templates for 2, 5, and 6 panels), computing bounding boxes from page width/height, margin, and gutter values so panels tile without manual placement.